

Lab 0

Aim: To study SQL commands, their uses, syntax, and install MySQL for database management experiments.

1. What is SQL and its Use?

SQL (Structured Query Language) is a standard language used to access and manipulate databases. It enables users to store, retrieve, update, and delete data in relational databases. SQL also provides commands to create and modify databases, tables, views, and other database objects. It became an ANSI standard in 1986 and an ISO standard in 1987 and is supported by database systems such as MySQL, SQL Server, Oracle etc.

SQL is used to execute queries against a database, retrieve data from database tables, insert new records, update existing records, and delete records when they are no longer needed. It is also used to create new databases and tables, create views and stored procedures, and manage permissions on tables, procedures, and views. These capabilities make SQL an essential language for storing, managing, and maintaining data in relational database management systems.

2. Describe all the different commands of SQL, mention use of it and syntax.

1.SELECT:

The SELECT command is used to retrieve data from one or more tables in a database.

Syntax:

```
SELECT column_name  
FROM table_name;
```

Example:

```
SELECT 'Ram'  
FROM Student;
```

2.UPDATE:

The UPDATE command is used to modify existing records in a table.

Syntax:

```
UPDATE table_name  
SET column_name = value  
WHERE condition;
```

Example:

```
UPDATE Student  
SET Name = 'Ram'  
WHERE Roll_No = 101;
```

3.DELETE:

The DELETE command is used to remove records from a table based on a specified condition.

Syntax:

```
DELETE FROM table_name  
WHERE condition;
```

Example:

```
DELETE FROM Student  
WHERE Name = 'Ram';
```

4.INSERT INTO:

The INSERT INTO command is used to add new records into a table.

Syntax:

```
INSERT INTO table_name (column1, column2)
VALUES (value1, value2);
```

Example:

```
INSERT INTO Student (Roll_No, Name)
VALUES (101, 'Ram');
```

5.CREATE DATABASE:

The CREATE DATABASE command is used to create a new database.

Syntax:

```
CREATE DATABASE database_name;
```

Example:

```
CREATE DATABASE College;
```

6.ALTER DATABASE:

The ALTER DATABASE command is used to modify the properties or settings of an existing database.

Syntax:

```
ALTER DATABASE database_name
MODIFY NAME = new_database_name;
```

Example:

```
ALTER DATABASE College
MODIFY NAME = University;
```

7.CREATE TABLE:

The CREATE TABLE command is used to create a new table by defining its columns and data types.

Syntax:

```
CREATE TABLE table_name (
column1 datatype,
column2 datatype
);
```

Example:

```
CREATE TABLE Student (
Roll_No INT,
Name VARCHAR(50)
);
```

8.ALTER TABLE:

The ALTER TABLE command is used to add, modify, or delete columns in an existing table.

Syntax:

```
ALTER TABLE table_name
ADD column_name datatype;
```

Example:

```
ALTER TABLE Student
ADD Email VARCHAR(50);
```

9.DROP TABLE:

The DROP TABLE command is used to permanently delete an existing table and all its data.

Syntax:

DROP TABLE table_name;

Example:

DROP TABLE Student;

10.CREATE INDEX:

The CREATE INDEX command is used to create an index on one or more columns of a table to improve the speed of data retrieval.

Syntax:

```
CREATE INDEX index_name  
ON table_name (column_name);
```

Example:

```
CREATE INDEX idx_student_name  
ON Student (Name);
```

DROP INDEX:

The DROP INDEX command is used to remove an existing index from a table.

Syntax:

```
DROP INDEX index_name;
```

Example:

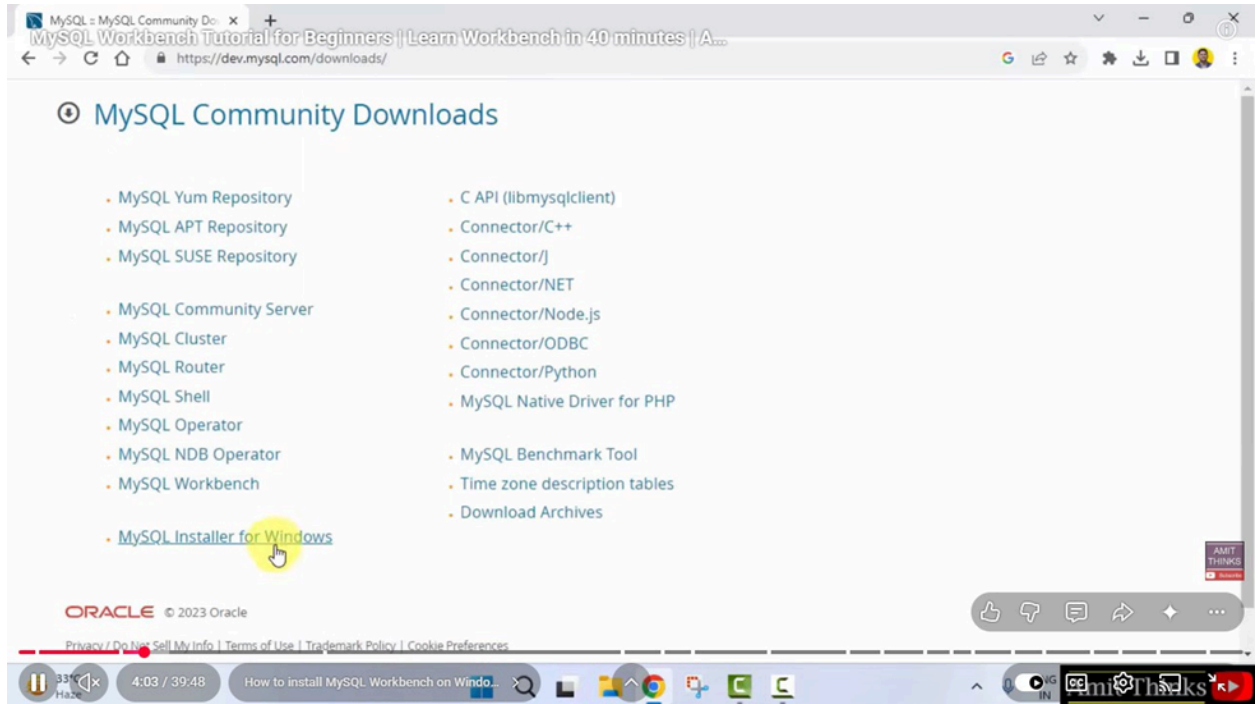
```
DROP INDEX idx_student_name;
```

3. Write step by step procedure to install MYSQL.

Download MySQL Installer

- Go to the official MySQL website.

- Navigate to **Downloads** → **MySQL Community Downloads** → **MySQL Installer for Windows**.
- Download the **MySQL Installer for Windows**.



2. Run the Installer

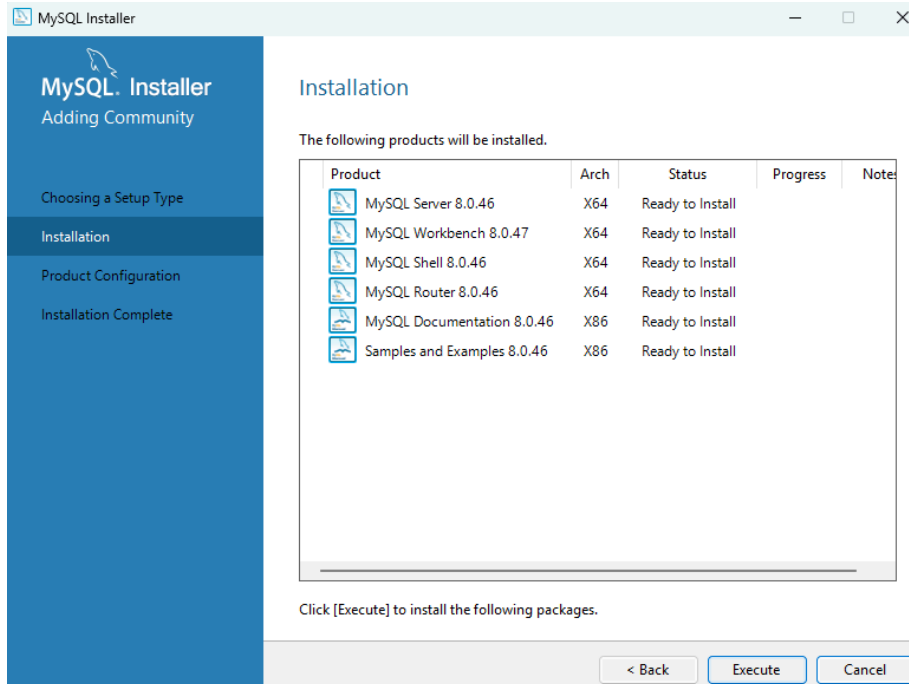
- Open the downloaded **.msi** file.
- If Windows asks for permission, click **Yes**.

3. Install Requirements

- The installer may show missing requirements such as **Visual C++ Redistributable**.
- Install the required components if prompted.
- Click **Next**.

4. Install MySQL Products

- Click **Execute** to install the selected components.
- Wait until all products show **Complete**.
- Click **Next**.

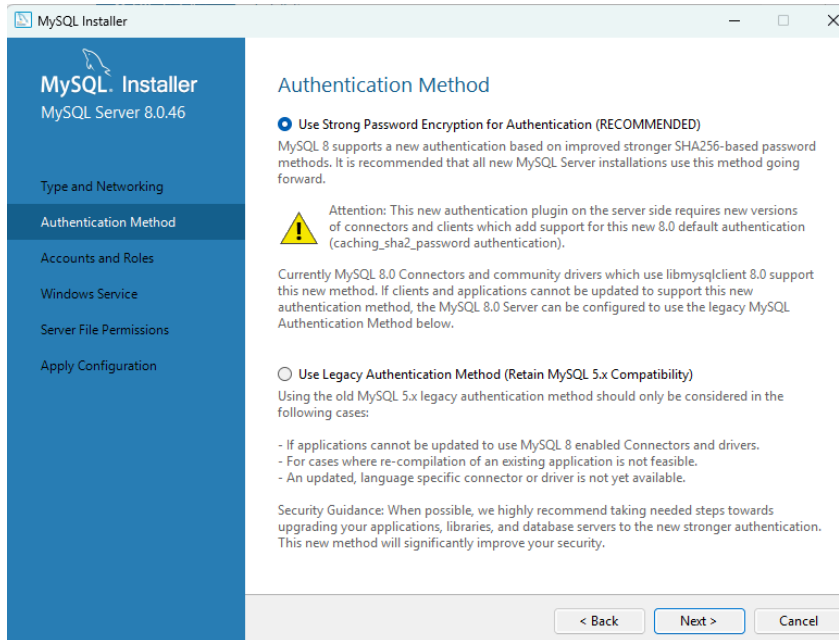


5. Configure MySQL Server

- Select **Standalone MySQL Server**.
- Keep the default configuration unless you have a specific requirement.
- Click **Next**.

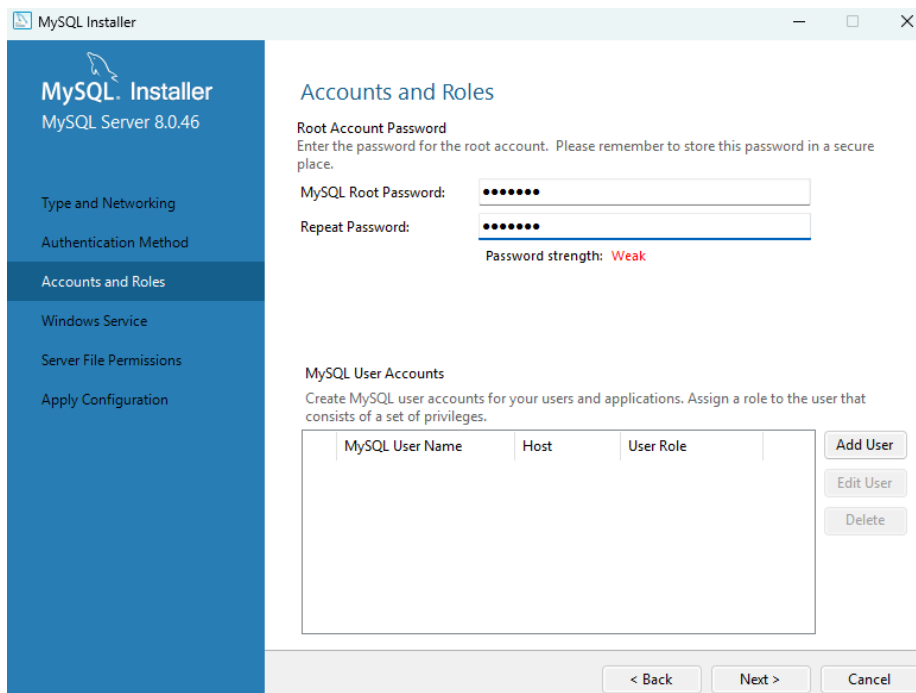
6. Set Authentication Method

- Keep the recommended/default authentication option.
- Click **Next**.



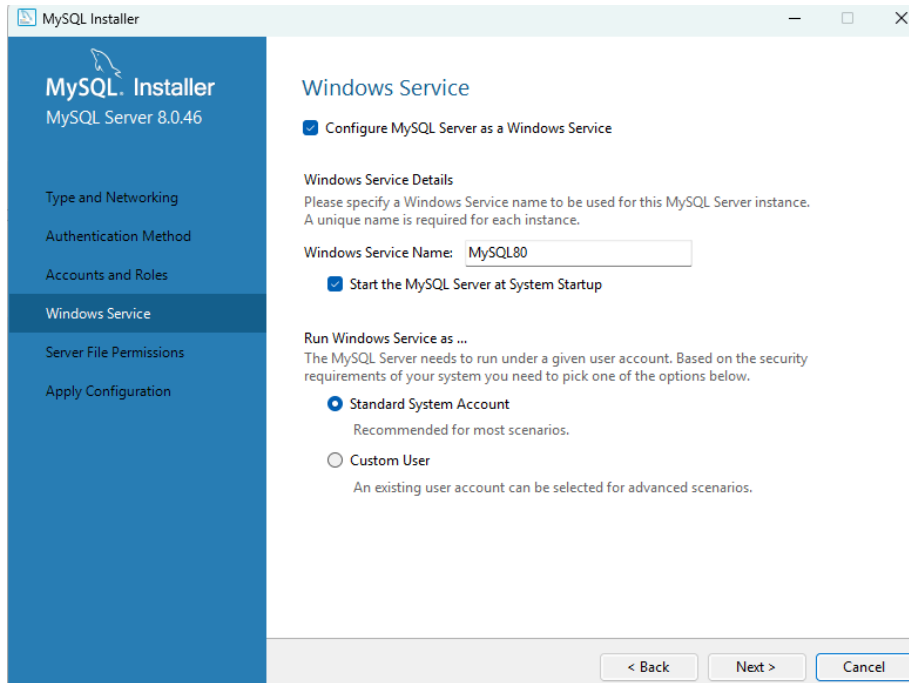
7. Set Root Password

- Enter a password for the MySQL root account.
- Re-enter it to confirm.
- **Remember this password**, because you will need it to log in to MySQL.



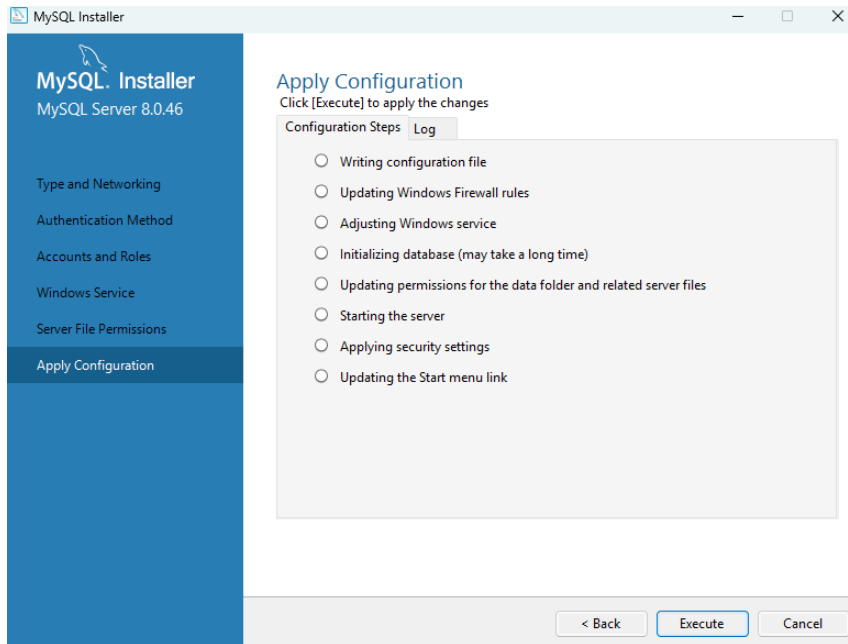
8. Configure Windows Service

- Keep **MySQL as a Windows Service** enabled.
- Leave the default service name unless required otherwise.
- Click **Next**.



10. Apply Configuration

- Click **Execute**.
- Wait for all configuration steps to complete.
- Click **Finish**.



11.Install MySQL Workbench

- If Workbench was included in your selected setup, it will be installed automatically.
- MySQL Workbench provides a graphical interface for creating databases, tables, and running SQL queries.

12.Verify Installation

- Open **MySQL Workbench**.
- Select your local MySQL connection.
- Enter the root password you created.
- Run: `SELECT VERSION();`
- If it returns the MySQL version, the installation is successful.

```

Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SELECT VERSION();
+-----+
| VERSION() |
+-----+
| 8.0.46    |
+-----+
1 row in set (0.00 sec)

mysql> |

```

MySQL localhost:3306 ssl pranav_dubey_d6_16 SQL > select *
from batting;

```

+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+

| MatchID | PID    | POrder | HOut | FOW | NRuns | Mts | Nballs |
Fours | Six |

+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+

|    2755 | 23001 |      3 | C    | 51  |      0 | 12  |      6 |
0 |    0 |

|    2755 | 25001 |      5 | C    | 232 |     71 | 104 |     80 |
4 |    0 |

```

[illegible][illegible][illegible][illegible][illegible]

2689	25001	5	C	240	36	52	37
2	1						

2689	99002	1	C	2	2	3	3
0	0						

2689	95001	3	C	8	1	9	7
0	0						

[illegible][illegible][illegible]

	2755		27001		9		B		257		2		7		6	
0		0														

	2689		98002		8		LBW		249		3		7		3	
0		0														

	2755		98002		8		R0		253		2		7		4	
0		0														

```

+-----+-----+-----+-----+-----+-----+-----+-----+
-----+-----+

```

16 rows in set (0.0005 sec)

MySQL localhost:3306 ssl pranav_dubey_d6_16 SQL > select *
from player;

```

+-----+-----+-----+-----+-----+-----+
-----+-----+

```

	PlayerID		LName		FName		Country		Yborn	
Bplace		FTest								

```

+-----+-----+-----+-----+-----+-----+
-----+-----+

```

	21001		Sehwag		Virender		India		1978	
Delhi		2001								

	23001		Singh		Yuvraj		India		1981	
Chandigarh		2003								

	24001	Symonds	Andrew	Australia	1975	
Birmingham		2004				
	25001	Dhoni	MS	India	1981	
Ranchi		2005				
	27001	Kumar	Praveen	India	1986	
Meerut		NULL				
	27002	Sharma	Ishant	India	1988	
Delhi		2007				
	89001	Tendulkar	Sachin	India	1973	
Mumbai		1989				
	90001	Lara	Brain	West Indies	1969	
Santa Cruz		1990				
	90002	Kumble	Anil	India	1970	
Bangalore		1990				
	91001	Jayasuriya	Sanath	Sri Lanka	1969	
Matara		1991				
	92001	Warne	Shane	Australia	1969	
Melbourne		1992				
	92002	Murlitharan	Muthiah	Sri Lanka	1972	
Kandy		1992				
	92003	UI-Huq	Inzamam	Pakistan	1970	
Multan		1992				

	93002	Streak	Heath	Zimbabwe	1974	
Bulawayo		1993				
	93003	Kirsten	Gary	South Africa	1967	
Cape Town		1993				
	94002	Vaas	Chaminda	Sri Lanka	1974	
Mattumagala		1994				
	94004	Fleming	Stephen	New Zealand	1973	
Christchurch		1994				
	95001	Ponting	Ricky	Australia	1974	
Launceston		1995				
	95002	Pollock	Shaun	South Africa	1973	
Port Elizabeth		1995				
	95003	Kallis	Jacques	South Africa	1975	
Cape Town		1995				
	96001	Dravid	Rahul	India	1973	
Indore		1996				
	96002	Gibbs	Herschelle	South Africa	1974	
Cape Town		1996				
	96003	Ganguly	Saurav	India	1972	
Calcutta		1996				
	97004	Vettori	Daniel	New Zealand	1979	
Auckland		1997				

	98001	Afridi	Shahid	Pakistan	1980	
Khyber Agency		1998				
	98002	Singh	Harbhajan	India	1980	
Jalandhar		1998				
	99001	Lee	Breet	Australia	1976	
Wollongong		1999				
	99002	Gilchrist	Adam	Australia	1971	
Bellingen		1999				
	99003	Vaughan	Michale	England	1974	
Manchester		1999				
	300311	Clark	Michale	NULL	NULL	
NULL		NULL				
	406317	NULL	Sam	India	NULL	
Ahmedabad		NULL				

```

+-----+-----+-----+-----+-----+
-----+-----+

```

31 rows in set (0.0005 sec)

MySQL localhost:3306 ssl pranav_dubey_d6_16 SQL > select *
from bowling;

```

+-----+-----+-----+-----+-----+-----+
| MatchID | PID   | NOvers | Maidens | Nruns | Nwickets |

```

+-----+-----+-----+-----+-----+-----+												
	2689		99001		10		0		58		1	
	2689		24001		3		0		27		1	
	2689		23001		3		0		15		0	
	2755		94002		9		1		40		1	
	2755		92002		10		0		56		1	
	2755		91001		4		0		29		0	
	2755		23001		10		0		53		2	
	2755		98002		10		0		40		3	
	2689		98002		10		0		44		1	

-----+-----+-----+-----+-----+-----+-----+

9 rows in set (0.0006 sec)

MySQL localhost:3306 ssl pranav_dubey_d6_16 SQL > show
tables;

-----+

| Tables_in_pranav_dubey_d6_16 |

-----+

| batsman |

| batting |

```
| bowling |
| bowling2689 |
| nplayer |
| odimatch |
| player |
```

```
+-----+
```

7 rows in set (0.0016 sec)

```
MySQL localhost:3306 ssl pranav_dubey_d6_16 SQL > select *
from odimatch;
```

```
+-----+-----+-----+-----+-----
--+
```

```
| MatchID | Team1 | Team2 | Ground | Date
| Winner |
```

```
+-----+-----+-----+-----+-----
--+
```

```
| 2324 | Pakistan | India | Peshwar |
2006-02-06 | Team1 |
```

```
| 2327 | Pakistan | India | Rawalpindi |
2006-02-11 | Team2 |
```

```
| 2357 | India | England | Delhi |
2006-03-28 | Team1 |
```

2377	West Indies	India	Kingston
2006-05-18	Team2		
2404	Sri Lanka	India	Colombo
2006-08-16	Abandoned		
2440	India	Australia	Mohali
2006-10-29	Team2		
2449	South Africa	India	Cape Town
2006-11-26	Team1		
2480	India	West Indies	Nagpur
2007-01-21	Team1		
2493	India	west Indies	Vadodara
2007-01-31	Team1		
2520	India	Sri Lanka	Rajkot
2007-02-11	Team2		
2611	England	India	Southampton
2007-08-21	Team1		
2632	India	Australia	Mumbai
2007-10-17	Team1		
2643	India	Pakistan	Guwahati
2007-11-05	Team1		
2675	Australia	India	Melbourne
2008-02-10	Team2		

2681	India	Sri Lanka	Adelaide
2008-02-19	Team1		
2688	Australia	India	Sydney
2008-03-02	Team2		
2689	Australia	India	Brisbane
2008-03-04	Team2		
2717	Pakistan	India	Karachi
2008-06-26	Team2		
2750	Sri Lanka	India	Colombo
2008-08-24	Team2		
2755	Sri Lanka	India	Colombo
2008-08-24	Team2		

```

+-----+-----+-----+-----+-----
--+-----+

```

20 rows in set (0.0004 sec)

```

MySQL localhost:3306 ssl pranav_dubey_d6_16 SQL > select *
from btasman;

```

ERROR: 1146 (42S02): Table 'pranav_dubey_d6_16.btasman' doesn't exist

```

MySQL localhost:3306 ssl pranav_dubey_d6_16 SQL > select *
from batsman;

```

```

+-----+-----+-----+-----+-----+-----+

```

PID	FName	LName	Country	MID	Score
23001	Yuvraj	Singh	India	2755	0
25001	MS	Dhoni	India	2755	71
91001	Sanath	Jayasuriya	Sri Lanka	2755	60
94002	Chaminda	Vaas	Sri Lanka	2755	17
92002	Muthiah	Murlitharan	Sri Lanka	2755	1
89001	Sachin	Tendulkar	India	2689	91
23001	Yuvraj	Singh	India	2689	38
25001	MS	Dhoni	India	2689	36
99002	Adam	Gilchrist	Australia	2689	2
95001	Ricky	Ponting	Australia	2689	1
24001	Andrew	Symonds	Australia	2689	42
99001	Breet	Lee	Australia	2689	7
27001	Praveen	Kumar	India	2689	7
27001	Praveen	Kumar	India	2755	2
98002	Harbhajan	Singh	India	2689	3
98002	Harbhajan	Singh	India	2755	2

+-----+-----+-----+-----+-----+-----+

16 rows in set (0.0006 sec)

MySQL localhost:3306 ssl prNAV_dubey_d6_16 SQL >